Storage Devices -- Guided Notes ANSWER KEY

Unit tp-2.3 | CompTIA Network+ N10-009 Obj. FC0-U71 2.3 | N10-008 FC0-U71 2.3

For instructor use / distribute after students complete the notes.

Question 1

A HDD (Hard Disk Drive) stores data on _______ platters that spin at speeds like 5,400 or 7,200 _______. A read/write _______ moves across the platters to access data. Because of these moving parts, HDDs are more susceptible to _______ damage from dropping.

Answer: Magnetic; RPM (revolutions per minute); head (arm); physical (shock).

Question 2

An SSD (Solid State Drive) stores data using _______ (NAND flash) chips with no moving parts. Compared to an HDD, an SSD is _______ (faster/slower), _______ (more/less) resistant to physical shock, and _______ (more/less) expensive per gigabyte.

Answer: Flash memory; faster; more resistant; more expensive.

Question 3

An M.2 NVMe SSD connects directly to the _______ bus via a slot on the motherboard, achieving speeds of _______ to _______ GB/s. A SATA SSD is limited to approximately _______ MB/s because SATA was designed for spinning hard drives.

Answer: PCIe; 3 to 7+ GB/s; 550-600 MB/s.

Question 4

SATA (Serial AT Attachment) is the interface used by most _______ and 2.5-inch _______ drives. SATA III has a maximum speed of _______ Gbps (approximately 600 MB/s). The SATA cable carries _______ only; a separate power cable connects from the PSU.

Answer: HDDs; SATA SSDs; 6 Gbps; data (signal).

Question 5

An optical drive reads and writes data to _______ using a laser. Common formats include CD (_______ MB), DVD (_______ GB), and Blu-ray (_______ GB). Optical drives are becoming less common in modern systems because _______ and digital distribution have replaced physical media.

Answer: Discs; 700 MB; 4.7 GB; 25-50 GB; USB drives (flash storage).

Question 6

RAID (Redundant Array of Independent Disks) combines multiple drives for performance or redundancy. RAID 0 (striping) _______ performance but provides _______ redundancy -- if one drive fails, all data is lost. RAID 1 (mirroring) duplicates data across _______ drives for redundancy at the cost of _______ the usable storage.

Answer: Improves; no; two; half (50%).

Question 7

NAS (Network Attached Storage) is a dedicated storage device connected to a network, allowing multiple _______ to access shared files. It differs from DAS (Direct Attached Storage) in that DAS connects _______ to a single computer, while NAS is accessible to anyone on the _______.

Answer: Clients (users/computers); directly; network.

Question 8 -- Real World Application

REAL WORLD: A school is upgrading 40 student laptops that currently use 256 GB HDDs. The budget allows for either SATA SSDs or NVMe SSDs as replacements. The primary use case is running Windows, a web browser, and Office applications. Which storage upgrade would you recommend and why? Is NVMe worth the additional cost for this scenario?

Answer: Recommendation: SATA SSD is the better choice for this scenario. Reasons: Both SATA SSD and NVMe SSD are dramatically faster than spinning HDDs for the workloads described (OS boot, browser, Office) -- students would see essentially identical real-world improvement from either. SATA SSDs cost significantly less per gigabyte than NVMe. The bottleneck for these tasks is not storage throughput -- it is RAM, CPU, and network latency. NVMe drives reach their advantage during large sequential reads/writes (video editing, virtual machines, large file transfers), which these students are not doing. Saving money on storage can fund other improvements like RAM upgrades.